Diego Felipe Peralta Peralta

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Java / Spring Boot Backend Developer | Full Stack & Applied AI

Computer Science Engineer specialized in Java, Spring Boot, and PostgreSQL. Experience developing REST APIs, multi-tenant SaaS platforms, and full-stack applications, integrating applied AI, security, automated testing, and cloud deployments. Focused on software architecture and maintainable solutions aligned with real business needs.

Education

Salesian Polytechnic University Cuenca, Ecuador

Computer Science Engineering. Studies completed. Degree in process.

Thesis: Design and implementation of an Active Visual Attention System and enhancement of the automatic speech recognition module for the Volky Robotic Assistant using deep learning networks, computer vision, and motion control.

Experience

ELECAUSTRO S.A Cuenca, Ecuador

Pre-professional Internship September 2025 - December 2025

- Developed a real-time SCADA monitoring system for PMSG wind turbines, integrating TCP/IP communication with MATLAB/Simulink, anomaly detection using Isolation Forest, and an interactive web interface in Streamlit built with a modular architecture (Python 3, scikit-learn, pandas).

Honors and Awards

Nomination: Outstanding Latin American Student Award 2026 — PENSER (Bothell, WA, USA).

Co-author: "Conceptual Modeling and Methodology for the Development of a Digital Twin for Wind Power Generation," IEEE GPECOM 2026, Naples, Italy. DOI: https://doi.org/10.1109/GPECOM70462.2026.11578476

Technical Projects

Medical Billing System — Clinical Management System | Full Stack in Production: Developed a full-stack clinical system for managing patients, appointments, medical records, billing, and payments, featuring business rules, state machines, and auditing. Integrated four AI features via RAG, vector search, Tool Calling, and structured outputs for clinical assistance, ICD-10 classification, and medical history queries. Backend Java 21 + Spring Boot 4, PostgreSQL/pgvector, Spring Security, JWT, and Redis; frontend React/TypeScript. Deployed on cloud infrastructure with automated tests. Demo: https://sistema-facturacion-medica-springbo.vercel.app

ClassroomIQ — AI-Assisted Teaching Evaluation Platform | Full Stack: Built a multi-tenant university evaluation support platform featuring RAG and draft generation with cited evidence, subject to professor review and approval. Implemented asynchronous document and code processing, semantic search, and similarity detection using PostgreSQL/pgvector, along with real-time updates via SSE. Backend Java 21 + Spring Boot 3.5, WebFlux, and Spring Security; frontend React 19 + TypeScript. Integration testing with Testcontainers and Playwright. Code: https://github.com/FepDev25/classroomiq

Ritmuvia — Multi-Tenant SaaS Platform for Gym Management | Full Stack & Cloud: Designed and developed a SaaS platform for gyms featuring membership management, payments, QR access, classes, reporting, and three user-role portals. Implemented multi-tenant isolation using PostgreSQL Row Level Security (RLS), granular permissions, and JWT authentication. FastAPI + React/TypeScript architecture with GitHub Actions CI/CD, deployed on Google Cloud Run, Neon, and Firebase Hosting. Documented suite of 665 tests (backend, frontend, and E2E), monitoring, backups, and disaster recovery procedures. Demo: https://ritmuvia.fepdev.app

Skills and Certifications

Languages and Frameworks: Java, Python, TypeScript, SQL, Spring Boot, Spring Security, Spring AI, FastAPI, React.

Databases: PostgreSQL, pgvector, Redis, SQLite, Flyway, Hibernate, Row Level Security (RLS).

Cloud, DevOps & Testing: Docker, Linux, Git, GitHub Actions, CI/CD, Google Cloud, Vercel, Cloudflare, JUnit, pytest, Testcontainers, Playwright.

AI & Architecture: RAG, LLMs, Embeddings, Tool Calling, REST APIs, multi-tenant design, software architecture, AI agents.

Certifications & Training*:* Frontend Masters (JavaScript, TypeScript, React/Next.js); Udemy (Python, Spring Boot, FastAPI).

Languages: Spanish (Native), English (B1)